

Frozen leakage-controlled rolling-origin evaluation design

Each connector links an expanding-window origin to its direct forecast target.

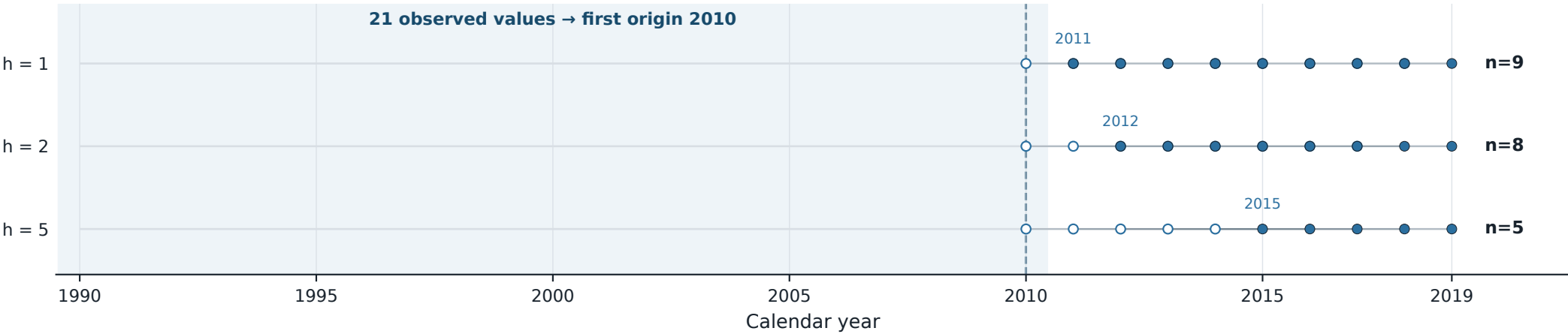
○ Forecast origin

● Forecast target

Initial 21-observation history

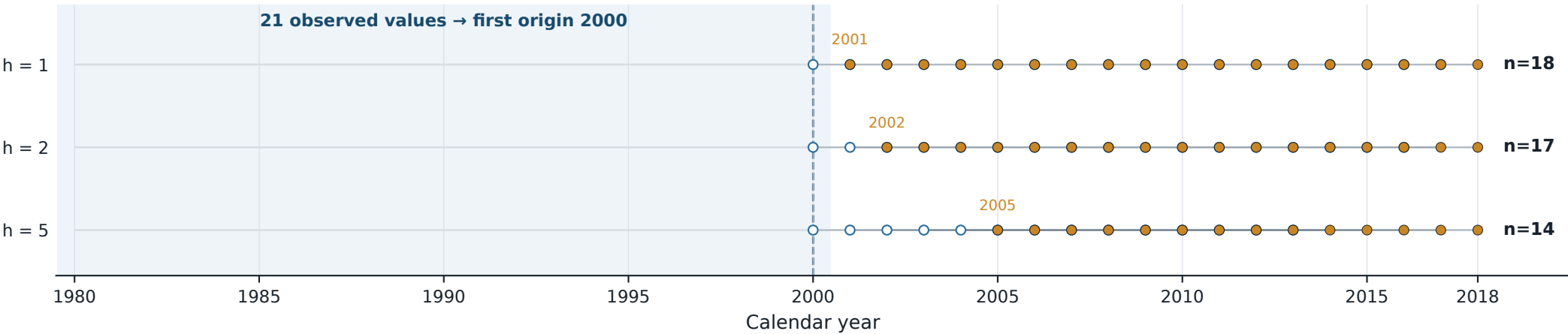
A

ASPH: official coverage 1990-2019



B

MTC: official coverage 1980-2018



C FROZEN PROTOCOL RULES

FIXED LOOKBACK

L = 3 for every primary ML and neural direct-horizon fit

SELECTION

Nested lookback selection disabled; legacy L = 5/6/8 not executed

EXCLUSIONS

No recent-period or regime analysis; no unsupported source years

LEAKAGE CONTROL

No observation at or after the target year enters fitting or preprocessing